SONG QIAOYUCHEN

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EDUCATION

National University of Singapore **Aug 2024 - Present**

Double Degree in Computer Science and Mathematics (GPA: 4.94/5.00 | Expected Graduation: May 2028)

- Relevant Coursework: Programming Methodology, Computer Organisation, Data Structures and Algorithms, Software Engineering, Database Systems, Computer Networks, Information Security, Design Thinking
- NUS Greyhats: Co-organised GreyCTF, Singapore's largest student-run CTF with 1000+ participants

SKILLS

- **Programming**: Java, C#, Python, C, C++, Go, JavaScript, TypeScript
- **Frameworks:** .NET, WPF, React, Next.js, Redux, FastAPI, Material UI, Tailwind CSS, Three.js
- **Databases**:PostgreSQL Microsoft SQL Server, Neo4j
- **Technologies**: MQTT, REST APIs, JWT, Supabase, Git

WORK EXPERIENCE

Sonicboom Solutions, *Software Engineer Intern* **May 2025 - Jul 2025**

- Redeveloped a C#/.NET WPF real-time monitoring console tracking parking-gantry status and operational events across 13 deployment sites.
- Implemented MQTT publish/subscribe messaging and asynchronous event handling between parking gantries and the monitoring console, enabling real-time alerts and status updates.
- Refactored WPF components and data-binding logic to improve maintainability and UI responsiveness for time-critical operator workflows.
- Partnered with system operators to identify usability problems and deliver UI/UX improvements.

PROJECTS

UniPlans | TypeScript, React, Next.js, Neo4j **May 2025 - Dec 2025**

- Founded and led the development of uniplans.net, a course planning platform that helps NUS students visualise degree requirements and generate informed academic plans under complex prerequisite constraints.
- Optimized prerequisite dependency queries by modeling complex academic relationships in Neo4j, reducing graph traversal time compared to relational lookups.
- Designed a constraint-aware greedy scheduling algorithm that generated feasible course plans based on prerequisites, workload preferences, and graduation timelines.
- Built an interactive drag-and-drop planner with real-time client-side validation, optimising frontend performance through normalised state, memoised selectors, and isolating the presentational layer to reduce re-render overhead at scale.
- Applied user-centred design principles to visualise complex prerequisite graphs in an intuitive and accessible interface.

Soforum | Go, PostgreSQL, React, Next.js, Redux, JWT **Dec 2024 - Jan 2025**

- Designed and built a full-stack web forum from scratch, supporting user discussions, thread management, and secure authentication with a clean MVC architecture.
- Developed a modern, responsive frontend using React and Material UI, leveraging Next.js for server-side rendering and Redux for predictable client-side state management.
- Implemented a RESTful backend in Go to handle user and thread operations, incorporating JWT-based authentication middleware to enforce secure access control.
- Integrated PostgreSQL for persistent data storage and deployed the application to a cloud environment using Render, configuring the system for production use.